Customer Churn Analysis Dashboard

Project Overview

Customer churn is one of the most critical challenges faced by telecom companies. Retaining existing customers is often more cost-effective than acquiring new ones. The objective of this project is to analyze customer behavior, identify factors contributing to customer churn, and provide actionable insights that can help improve customer retention.

This project uses the Telco Customer Churn dataset and applies data cleaning, SQL analysis, and Power BI visualization techniques to uncover churn patterns and business insights.

Objectives

- Analyze customer churn behavior.
- Identify key factors influencing customer attrition.
- Measure churn rate and customer retention metrics.
- Explore the impact of contract type, internet service, payment methods, and customer demographics on churn.
- Build an interactive Power BI dashboard for business decision-making.

Dataset Information

Dataset Name: Telco Customer Churn Dataset

Total Records: 7,000+

Total Columns: 33

Key Features:

- Customer Demographics
- Service Information
- Contract Details
- Billing Information
- Customer Lifetime Value (CLTV)
- Churn Status and Churn Reason

Target Variable:

- Churn Label (Yes/No)

Tools and Technologies Used

- Microsoft Excel – Data Cleaning and Preparation
- MySQL – Data Analysis and Querying
- Power BI – Dashboard Development and Visualization

Data Cleaning Process

The dataset was cleaned and prepared using Microsoft Excel. The following steps were performed:

- Checked for duplicate customer records.
- Verified missing and blank values.
- Validated numerical fields such as Monthly Charges and Total Charges.
- Standardized categorical values.
- Ensured consistency in Yes/No fields.
- Removed unnecessary inconsistencies before importing data into MySQL.

SQL Analysis

The cleaned dataset was imported into MySQL for analysis.

The following business questions were answered:

- Total number of churned customers.

```
select count(*) as churned_customers
from telco_customer_churn_clean
where `Churn Label`='Yes';
```

	churned_customers
▶	1869

- Customer churn rate.

```
select round(sum(case when `churn label`='Yes' then 1 else 0 end)*100/count(*),2) as ChurnRate
from telco_customer_churn_clean;
```

	ChurnRate
▶	26.58

- Churn distribution by gender.

```
select Gender,count(*) as ChurnbyGender
from telco_customer_churn_clean
where `Churn Label`='Yes'
group by Gender
order by ChurnbyGender desc ;
```

	Gender	ChurnbyGender
▶	Female	939
	Male	930

- **Churn distribution by contract type.**

```sql
select Contract,count(*) as Churned_customer
from telco_customer_churn_clean
where `Churn Label`='Yes'
group by Contract
order by Churned_customer desc ;
```

Contract	Churned_customer
Month-to-month	1655
One year	166
Two year	48

- **Churn distribution by internet service.**

```sql
select `Internet Service`,count(*) as Churned_customer
from telco_customer_churn_clean
where `Churn Label`='Yes'
group by `Internet Service`
order by Churned_customer desc ;
```

Internet Service	Churned_customer
Fiber optic	1297
DSL	459
No	113

- **Top reasons for customer churn.**

```sql
select `Churn Reason`,count(*) as Total_customer
from telco_customer_churn_clean
where `Churn Label`='Yes'
group by `Churn Reason`
order by Total_customer desc limit 10 ;
```

Churn Reason	Total_customer
Attitude of support person	192
Competitor offered higher download speeds	189
Competitor offered more data	162
Don't know	154
Competitor made better offer	140
Attitude of service provider	135
Competitor had better devices	130
Network reliability	103
Product dissatisfaction	102
Price too high	98

- **Average monthly charges by churn status.**

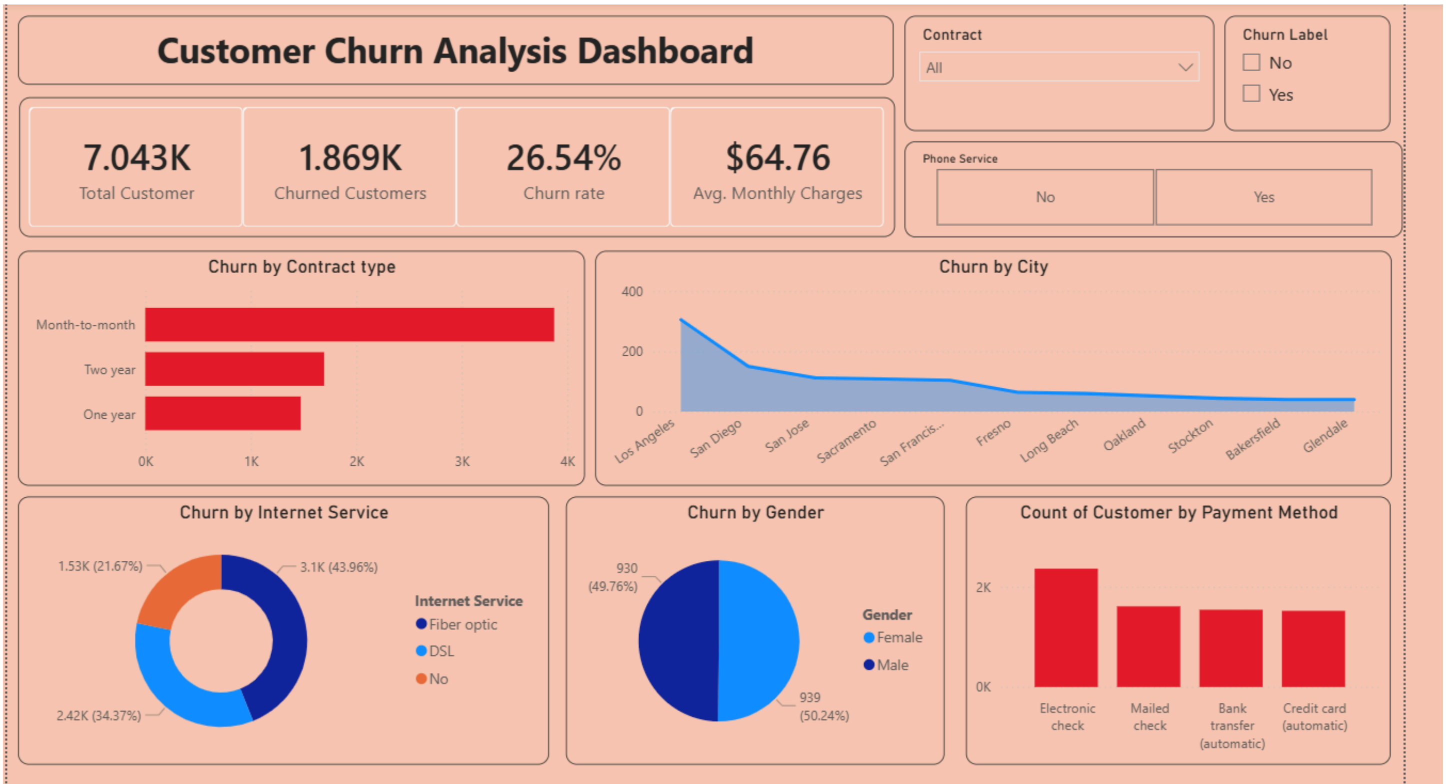
```sql
select `Churn Label`,round(avg(`Monthly Charges`),2) as Average_Monthlycharges
from telco_customer_churn_clean
group by `Churn Label`;
```

	Churn Label	Average_Monthlycharges
▶	Yes	74.44
	No	61.31

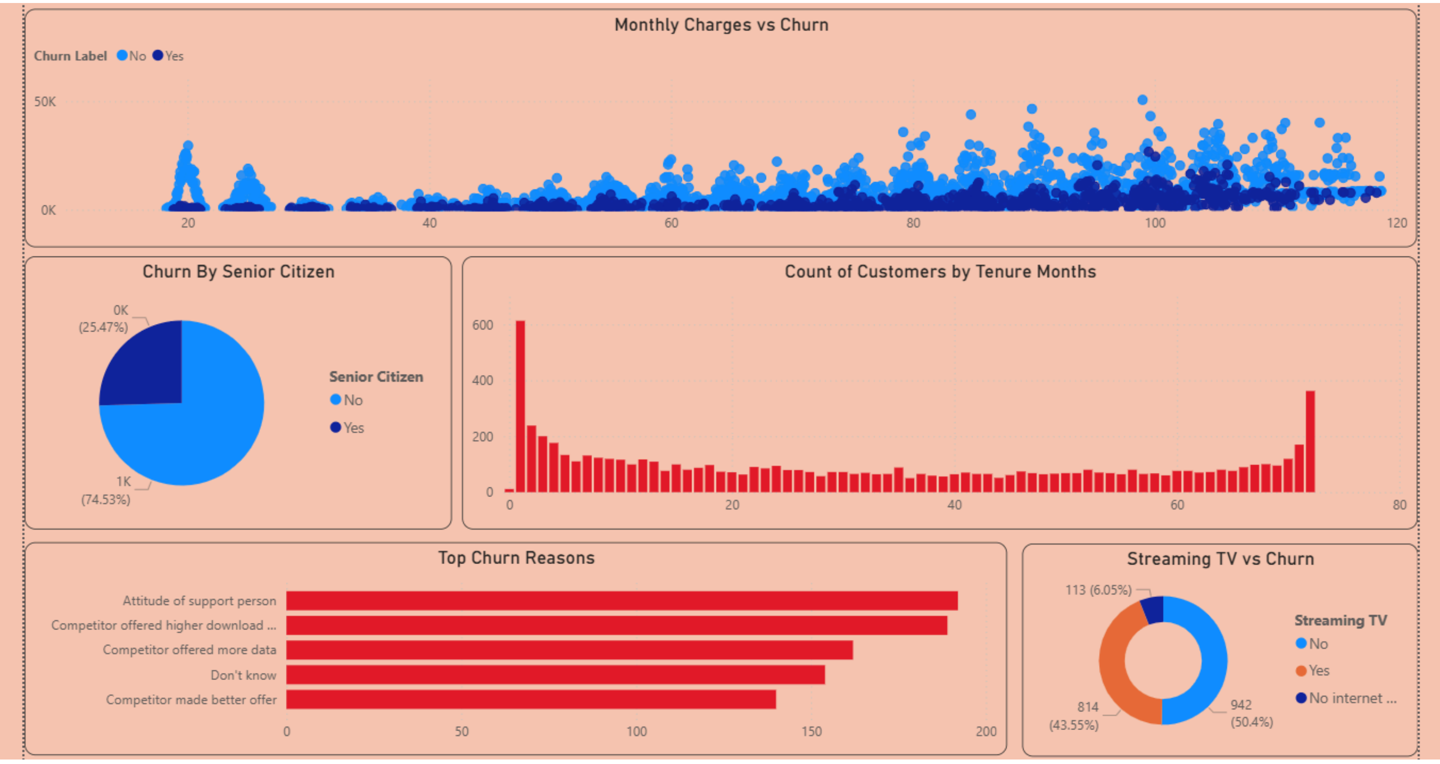
Power BI Dashboard

An interactive Power BI dashboard was created to visualize customer churn patterns and business metrics.

Dashboard Page 1: Executive Overview



Dashboard Page 2: Customer Insights



Key Findings

- Customers on month-to-month contracts exhibit significantly higher churn rates compared to customers with longer-term contracts.
- Customers using fiber optic internet services show a higher tendency to churn.
- Higher monthly charges are associated with increased churn probability.
- Certain payment methods contribute to higher churn rates.
- Customer tenure plays a significant role in retention, with newer customers being more likely to leave.
- Churn reasons indicate that pricing, competition, and service-related concerns are major drivers of customer attrition.

Conclusion

The Customer Churn Analysis project successfully identified the major factors contributing to customer attrition. Contract type, monthly charges, customer tenure, and internet service preferences were found to have a strong impact on churn behavior.

The insights generated through SQL analysis and Power BI dashboards can help telecom companies develop targeted retention strategies, improve customer satisfaction, and reduce customer loss.

This project demonstrates practical skills in data cleaning, SQL querying, business analysis, and dashboard development using Power BI.